

For the week of September 15-19

- 1) Evaluate the following integrals by simplifying and using the antiderivatives of elementary functions.

a) $\int \frac{2^{x+1} - 5^{x-1}}{10^x} dx.$

b) $\int \frac{3e^x x^3 - 6x^2 + x}{3x^3} dx.$

c) $\int \left(1 - \frac{1}{x^2}\right) \sqrt{x} \sqrt{x} dx.$

d) $\int \frac{x^3 - 3x^2 + 3x - 1}{x - 1} dx.$

e) $\int \left(\frac{1-x}{x}\right)^2 dx.$

f) $\int \sin 5x - \sin 5 dx.$

g) $\int \frac{e^{3x} + 1}{e^x + 1} dx.$

h) $\int (1 + \tan^2 x) \sqrt{1 - \cos^2 x} dx.$

i) $\int \frac{x^3 + 6x^2 + 11x + 6}{x - 1} dx.$

- 2) For each pair of functions below, determine the total area of the region bound between them.

a) $f(x) = (2x)^2$ and $g(x) = \sqrt{2x}.$

b) $f(x) = |2x|$ and $g(x) = x + 2.$

c) $f(x) = \sin 2x$ and $g(x) = \cos 2x$ on $\left[0, \frac{\pi}{2}\right]$

- 3) Determine the tangent line approximation of the integral function

$$f(x) = \int_0^x \frac{\sin t}{t} dt$$

about the point $x = 0.$

- 4) Let f be a continuous function and let

$$F(x) = \int_{1-x}^{1+x^3} f(t) dt.$$

- a) If possible, find a number $c \in \mathbb{R}$ such that $F(c) = 0$. If it is not possible, explain why.
- b) Show that the average value of $f(t)$ on $[0, 2]$ is $\frac{1}{2}F(1)$.
- c) For this part only, let $f(t) = \ln(t^2 + 1)$. Find $F'(x)$.

- 5) For each function F below, find all of its critical points.

a) $F(x) = \int_{-1}^x t^{10} e^t dt.$

b) $F(x) = \int_2^x e^{-t^2} - 1 dt.$

c) $F(x) = \int_{x^2}^1 t - 1 dt.$

d) $F(x) = \int_0^{\sin x} dt.$

- 6) The acceleration due to gravity—at least on earth—is about -10m/s^2 . You stand at the edge of a bottomless pit and throw a ball upward. Its initial velocity is 25m/s .

- a) Find the average velocity of the ball in the first 5 seconds of its travel.
- b) Find the total distance travelled by the ball in the first 5 seconds of its travel.

- 7) Suppose that $f'(x^2) = \frac{1}{x}$ ($x > 0$). Find $f(x)$.

- 8) For each function f below, find the antiderivative F whose graph passes through the indicated point.

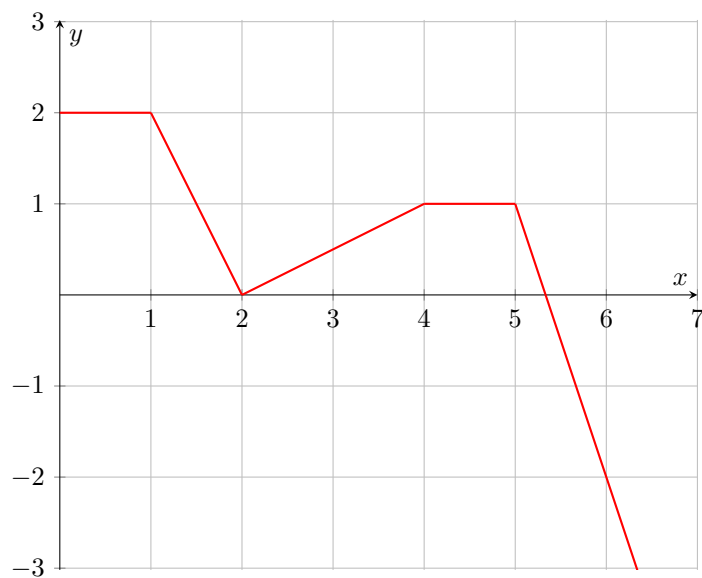
a) $f(x) = \frac{x+1}{\sqrt{x}}$, $F(x)$ passes through $(4, 1)$.

b) $f(x) = (2^x + 3^x)^2$, $F(x)$ passes through $(0, 0)$.

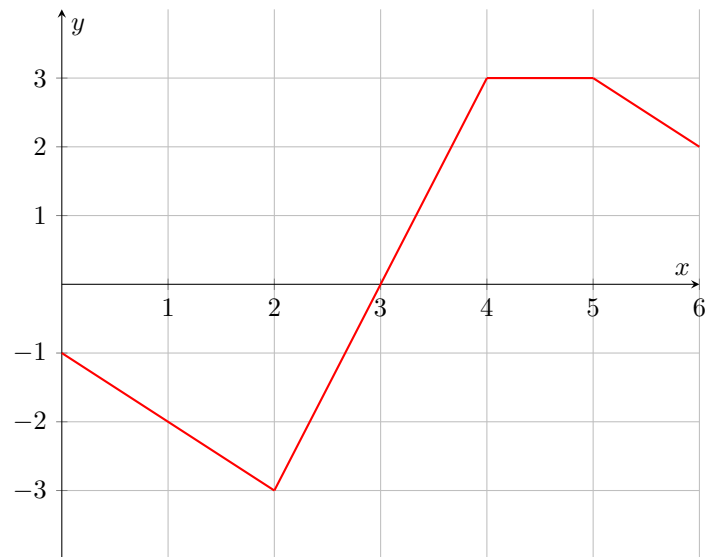
c) $f(x) = \frac{x-1}{x^2-x}$, $F(x)$ passes through $(1, 1)$.

- 9) For each part below, you are given the graph of a function f and the value of its antiderivative F at some point. Determine the value of its antiderivative at the specified point. The graphs are composed of straight lines and circular segments.

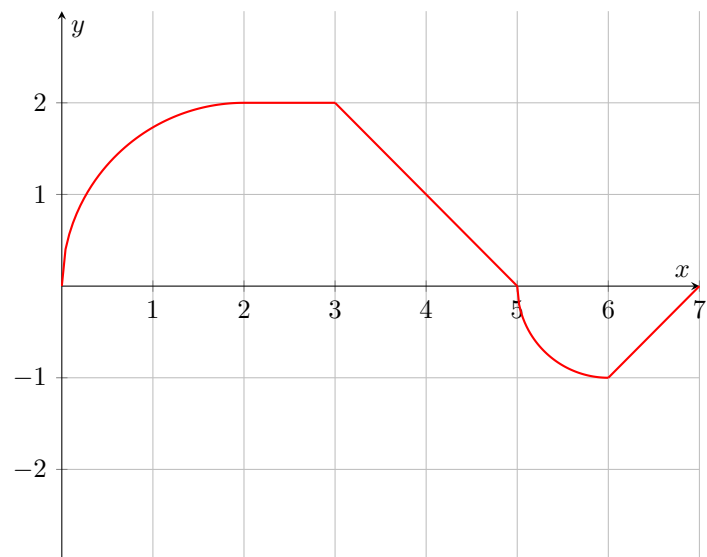
a) Find $F(1)$ given that $F(5) = 5$.



b) Find $F(5)$ given that $F(0) = 1$.



c) Find $F(2)$ given that $F(6) = 0$.



The following problems are for practice only and will not appear on quizzes.

- I) Textbook section 6.2, problems # 6-56, 81-96.
- II) Textbook section 6.3, problems # 3-15, 17-19, 26-28.
- III) Textbook section 6.4, problems # 11-22, 28, 30, 31, 38-45, 53-58.